

02. Geometric Camera Models

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Outline



- Image Transformation
- Geometric Camera Model

Factors in image formation

■ **Geometry**

- Relationship between points in the 3D world and their images

■ **Radiometry**

- Relationship between the amount of light radiating from a surface and the amount of incident at its image
- Measurement of optical radiation (ultraviolet, visible and infrared)

■ **Photometry**

- Ways of measuring the intensity of light
- Measurement of light that is detectable by human eye

■ **Digitization**

- Ways of converting continuous signals to digital approximations

Image Transformation

Image Transformations

What is the geometric relationship between these two images?



Image Transformations

What is the geometric relationship between these two images?



Why don't these images line up exactly?

Image Transformations

What is the geometric relationship between these two images?

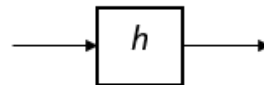
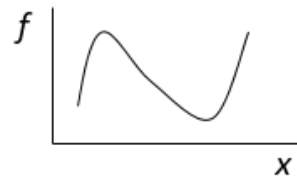


Image Transformations

Image Warping

- image filtering: change *range* of image

$$g(x) = h(f(x))$$



- image warping: change *domain* of image

$$g(x) = f(h(x))$$

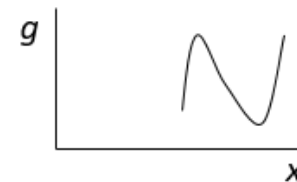
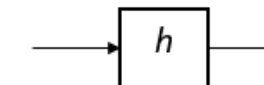


Image Transformations

Geometric primitives

■ *2D points*

- $x = (x, y) \in R^2$

■ *2D lines*

- $\bar{x} \cdot \bar{l} = ax + by + c = 0$
- $\bar{l} = (a, b, c)$

■ *3D points*

- $\bar{x} = (\bar{x}, \bar{y}, \bar{z}, \bar{w})$

■ *3D planes*

- $\bar{x} \cdot \bar{m} = ax + by + cz + d = 0$
- $\bar{m} = (a, b, c, d)$

■ *3D lines*

- $r = (1 - \lambda)p + \lambda q$

Image Transformations

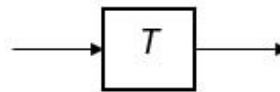
Parametric (global) warping

- Transformation T is a coordinate-changing machine:

$$\mathbf{p}' = T(\mathbf{p})$$



$\mathbf{p} = (x, y)$



$\mathbf{p}' = (x', y')$

- Let's consider *linear* xforms (can be represented by a 2x2 matrix) :

$$\mathbf{p}' = \mathbf{T}\mathbf{p}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \mathbf{T} \begin{bmatrix} x \\ y \end{bmatrix}$$

Image Transformations

Common linear transformations

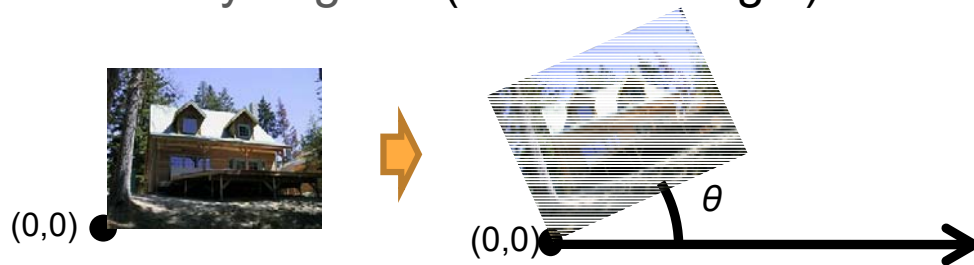
- Uniform scaling by s :



$$\mathbf{S} = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix}$$

What is the inverse?

- Rotation by angle θ (about the origin) :



$$\mathbf{R} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Image Transformations

2x2 Matrices

- 2D mirror about Y axis?

$$\begin{aligned}x' &= -x \\ y' &= y\end{aligned}\quad \mathbf{T} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

- 2D mirror across line $y = x$?

$$\begin{aligned}x' &= y \\ y' &= x\end{aligned}\quad \mathbf{T} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Image Transformations

2x2 Matrices

- 2D Translation?

$$\begin{aligned}x' &= x + t_x \\ y' &= y + t_y\end{aligned}$$

$$\mathbf{T} = \left[\begin{array}{c} \text{?} \end{array} \right]$$

Translation is not a linear operation on 2D coordinates

Image Transformations

All 2D Linear Transformations

- Linear transformations are combinations of ...

- Scale,
- Rotation,
- Shear, and
- Mirror

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

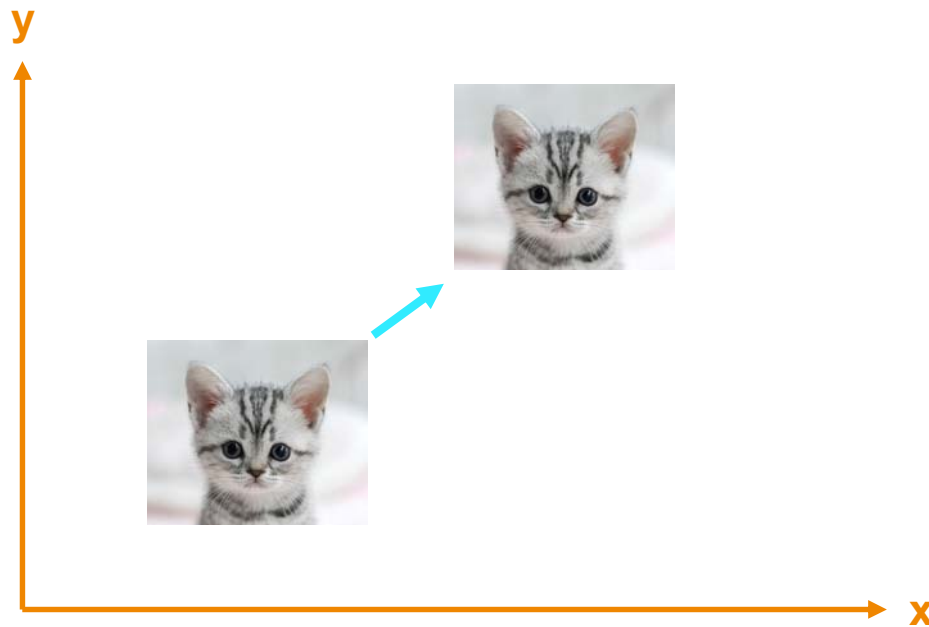
- Properties of linear transformations:

- Origin maps to origin
- Lines map to lines
- Parallel lines remain parallel
- Ratios are preserved
- Closed under composition

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} \begin{bmatrix} i & j \\ k & l \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Image Transformations

Homogeneous Coordinates



$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} ? & ? \\ ? & ? \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{aligned} x' &= x + t_x \\ y' &= y + t_y \end{aligned}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Heterogeneous
coordinates

Homogeneous
coordinates

$$\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x + t_x \\ y + t_y \\ 1 \end{bmatrix}$$

Image Transformations

Homogeneous Coordinates

Trick: add one more coordinate:

$$(x, y) \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad \text{homogeneous image coordinates}$$

Converting *from* homogeneous coordinates

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} \Rightarrow (x/w, y/w)$$

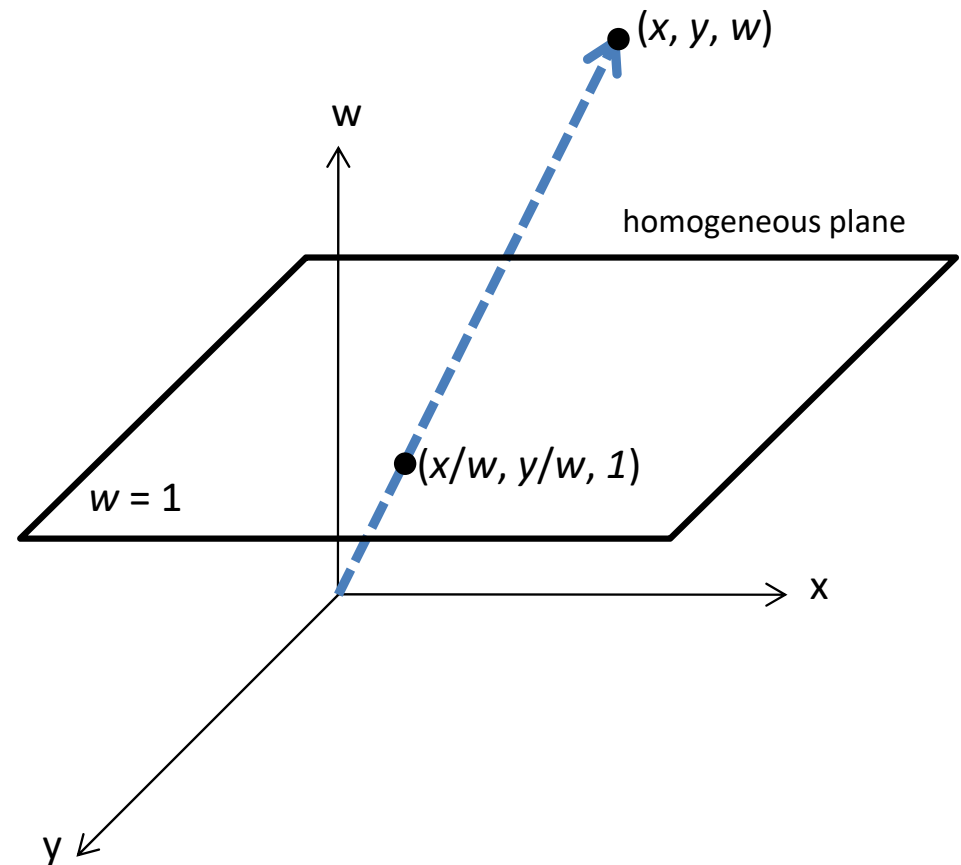


Image Transformations

- Rigid Transformation
- Similarity Transformation
- Affine Transformation
- Perspective Transformation

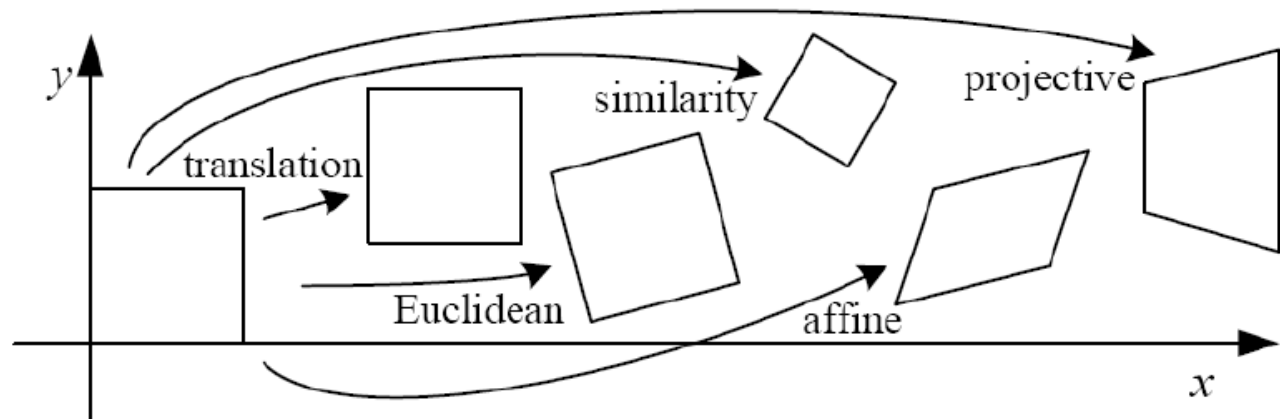
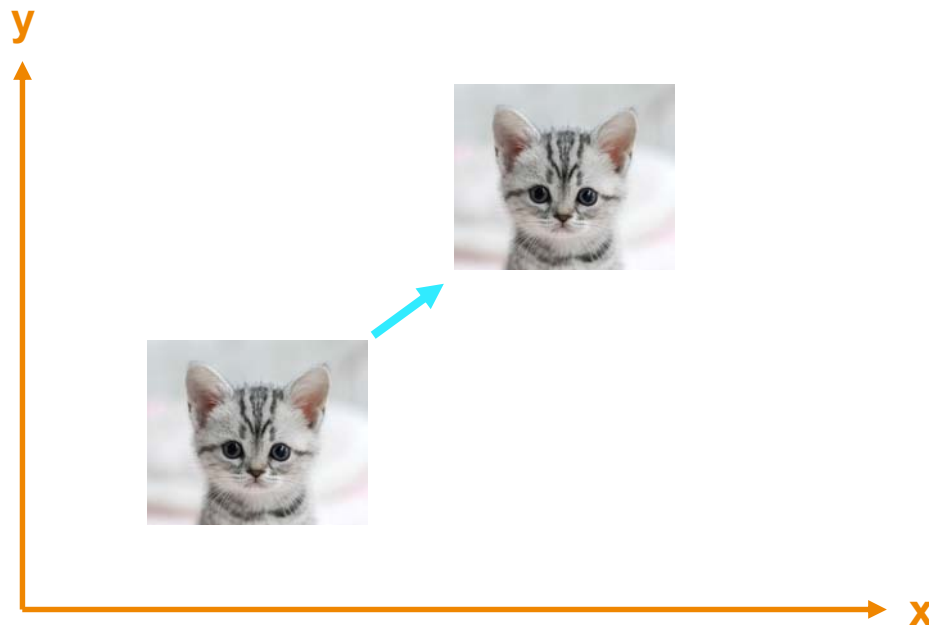


Image Transformations

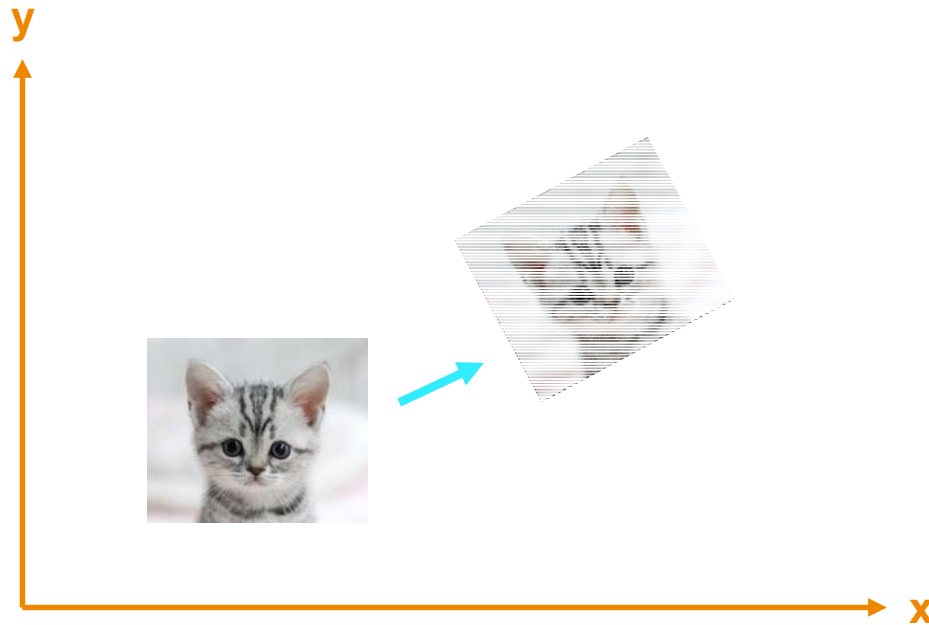
Rigid Transformation - Translation



$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x + t_x \\ y + t_y \\ 1 \end{bmatrix}$$

Image Transformations

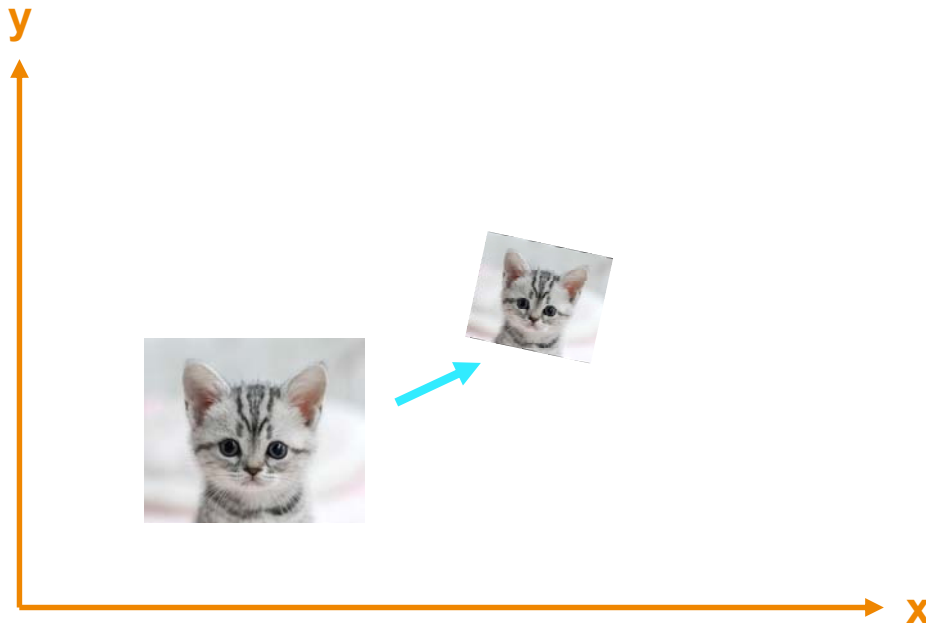
Rigid Transformation - Euclidean



$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & t_x \\ \sin \theta & \cos \theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Image Transformations

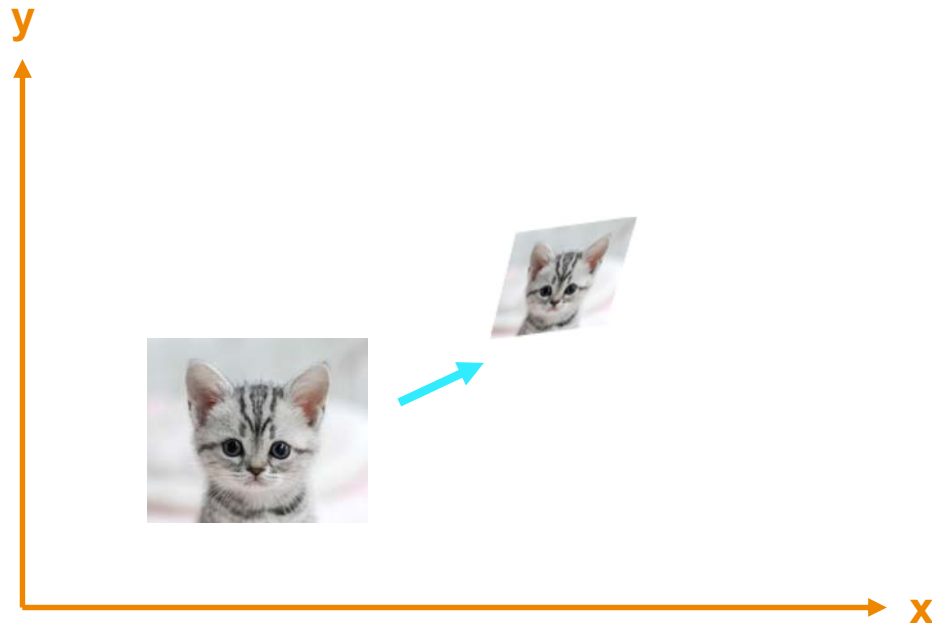
Similarity Transformation



$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} a & -b & e \\ b & a & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Image Transformations

Affine Transformation



$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} a & b & e \\ c & d & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Image Transformations

Affine Transformation

- Affine transformations are combinations of ...
 - Linear transformations, and
 - Translations

- Properties of affine transformations:
 - Origin does not necessarily map to origin
 - Lines map to lines
 - Parallel lines remain parallel
 - Ratios are preserved
 - Closed under composition

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} a & b & e \\ c & d & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Image Transformations

Affine Transformation

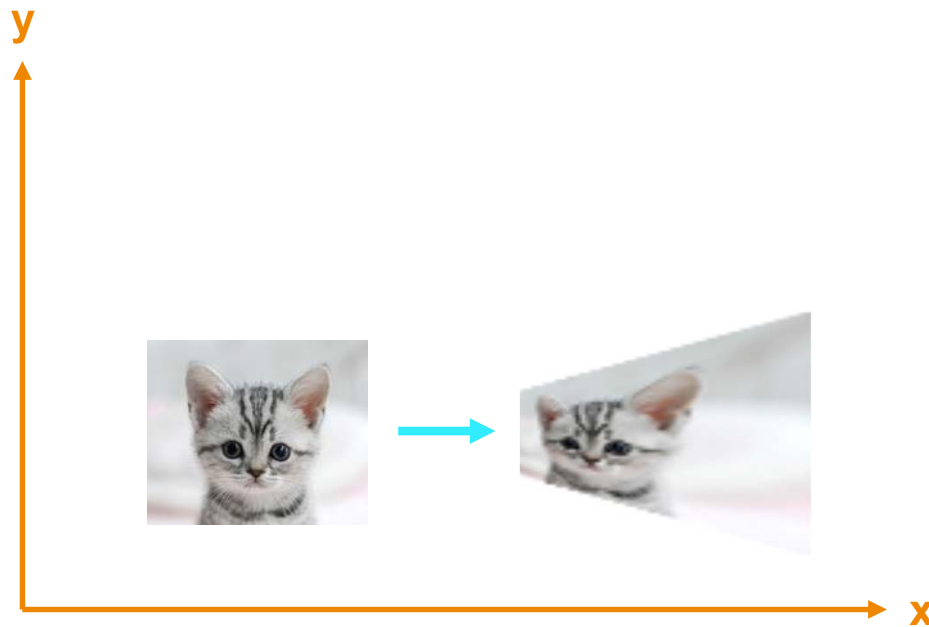


$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} a & b & e \\ c & d & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad ?$$

Is this an affine transformation?

Image Transformations

Perspective Transformation



$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} a & b & e \\ c & d & f \\ g & h & i \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$



Image Transformations

Perspective Transformation

- Homographies ...
 - Affine transformations, and
 - Projective warps

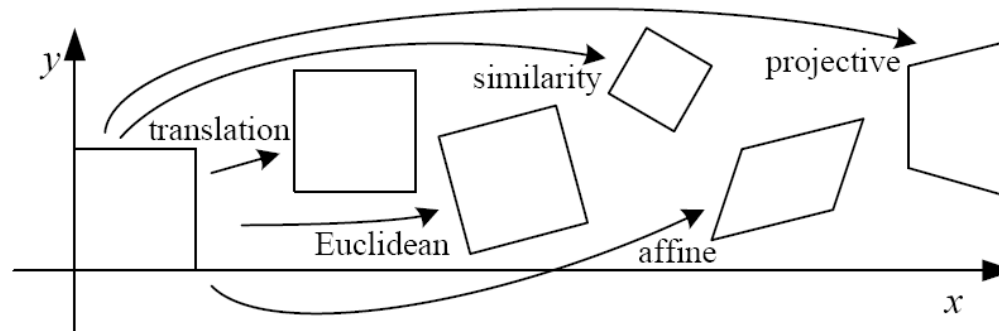
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} a & b & e \\ c & d & f \\ g & h & i \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- Properties of projective transformations:
 - Origin does not necessarily map to origin
 - Lines map to lines
 - Parallel lines do not necessarily remain parallel
 - Ratios are not preserved
 - Closed under composition

Image Transformations

https://opencv-python-tutroals.readthedocs.io/en/latest/py_tutorials/py_imgproc/py_geometric_transformations/py_geometric_transformations.html

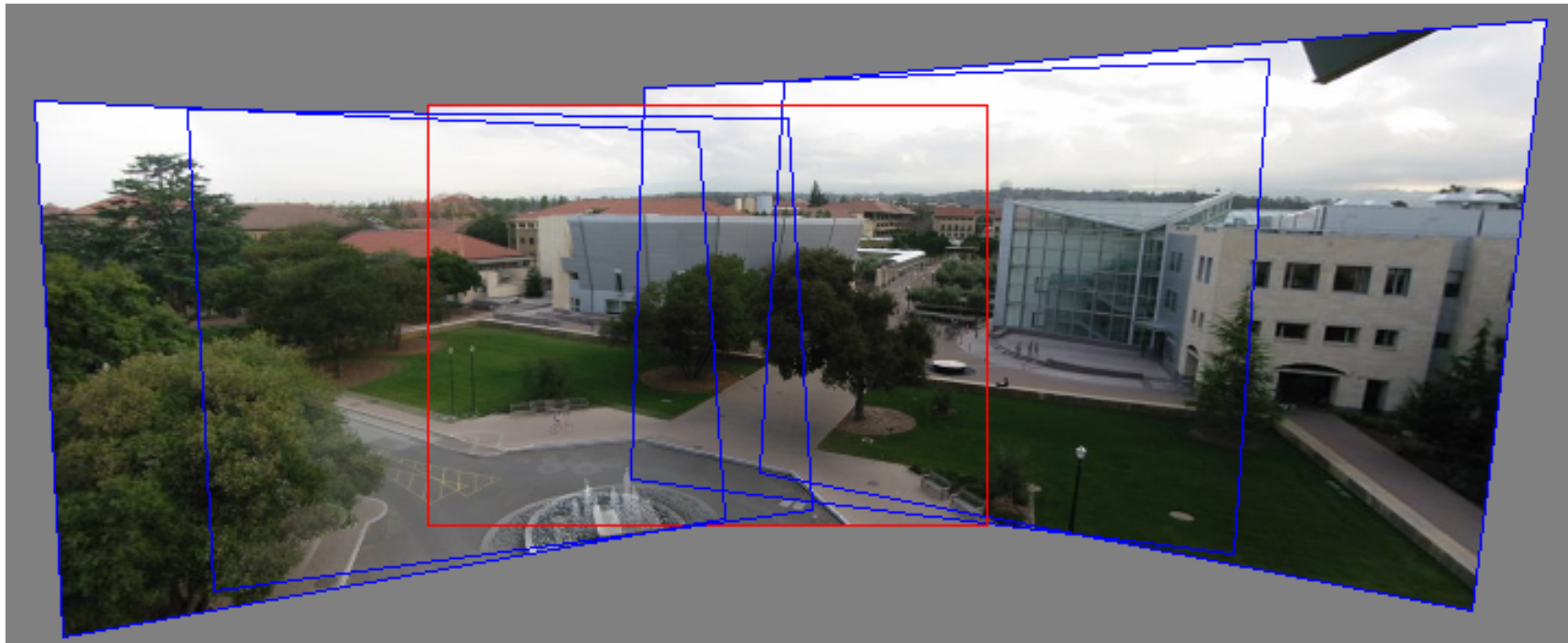
■ 2D transformations



Name	Matrix	# D.O.F.	Preserves:	Icon
translation	$\begin{bmatrix} I & t \end{bmatrix}_{2 \times 3}$	2	orientation + ...	
rigid (Euclidean)	$\begin{bmatrix} R & t \end{bmatrix}_{2 \times 3}$	3	lengths + ...	
similarity	$\begin{bmatrix} sR & t \end{bmatrix}_{2 \times 3}$	4	angles + ...	
affine	$\begin{bmatrix} A \end{bmatrix}_{2 \times 3}$	6	parallelism + ...	
projective	$\begin{bmatrix} \tilde{H} \end{bmatrix}_{3 \times 3}$	8	straight lines	

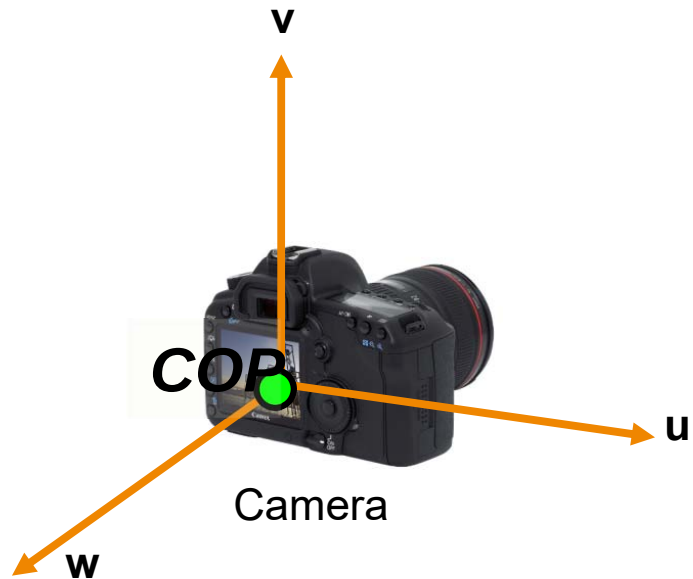
Geometric Camera Models

camera model



Camera parameters

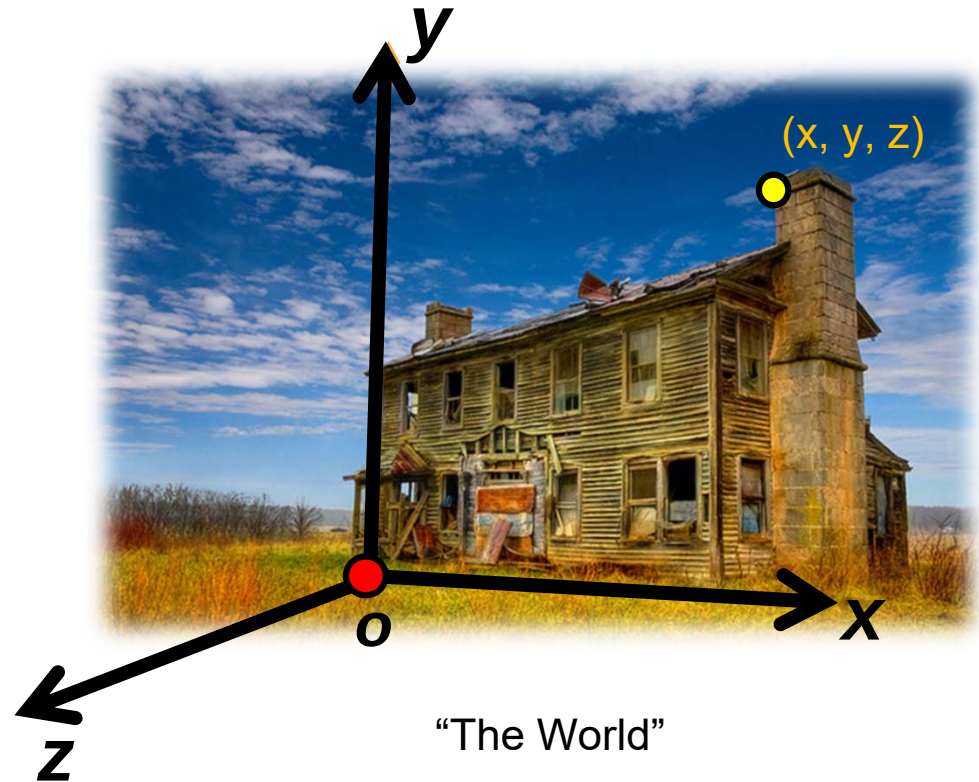
- How can we model the geometry of a camera?



Two important coordinate systems:

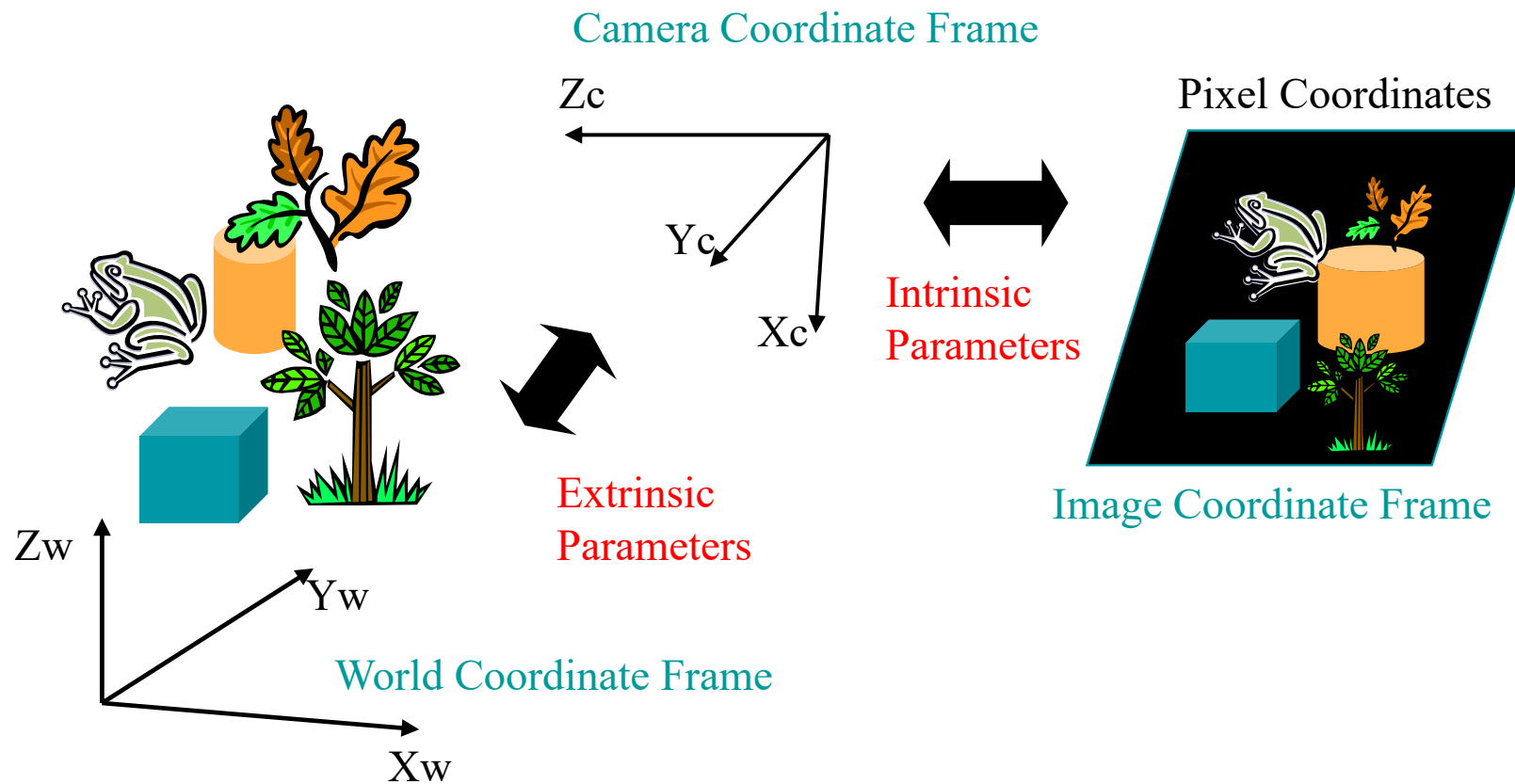
1. *World* coordinate system
2. *Camera* coordinate system

How do we project a given point (x, y, z) in world coordinates?



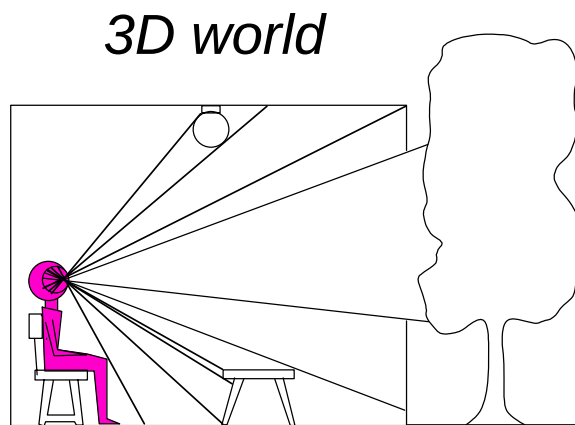
Camera parameters

- Camera calibration

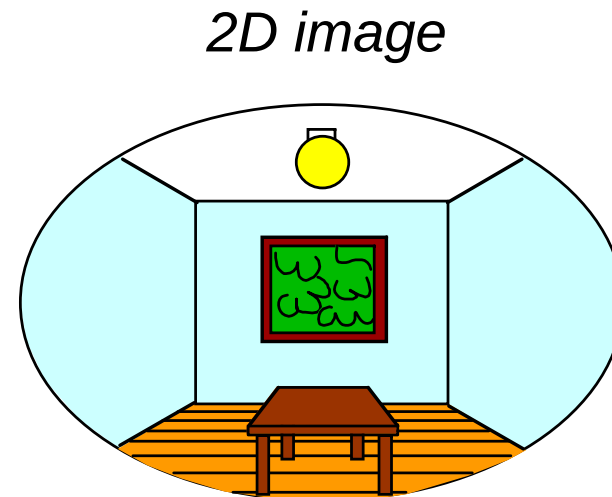


Camera parameters

- Dimensionality Reduction Machine (3D to 2D)
 - What have we lost? : Angles, Distances (lengths)
 - What have we preserved? : Straight lines, incidence



Point of observation



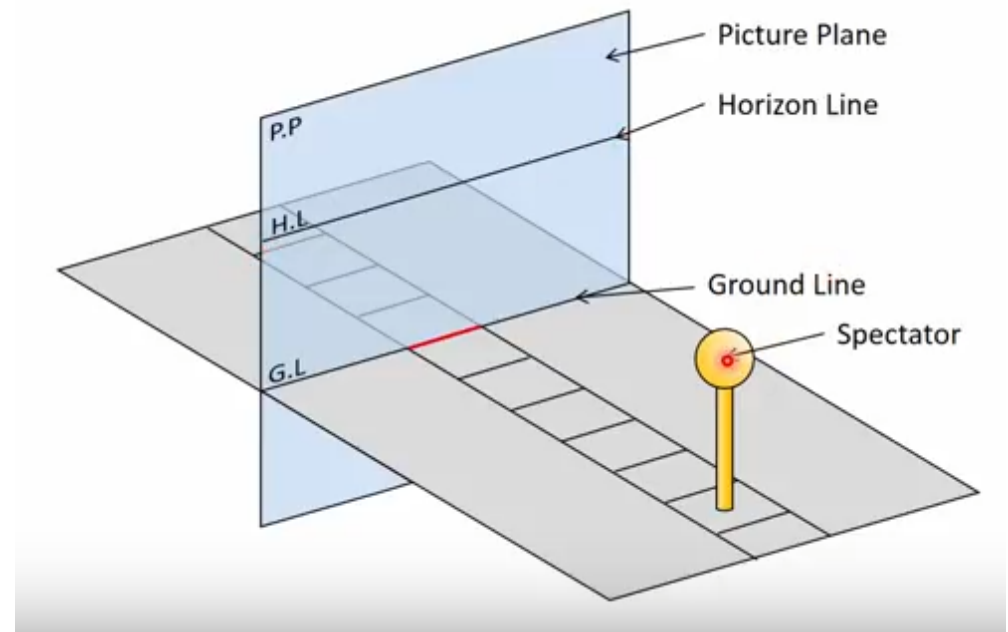
Figures © Stephen E. Palmer, 2002

Projection properties

- Many-to-one: any points along same ray map to same point in image
- Points $\rightarrow$ points
- Lines $\rightarrow$ lines (collinearity is preserved)
 - But line through focal point projects to a point
- Planes $\rightarrow$ planes (or half-planes)
 - But plane through focal point projects to line

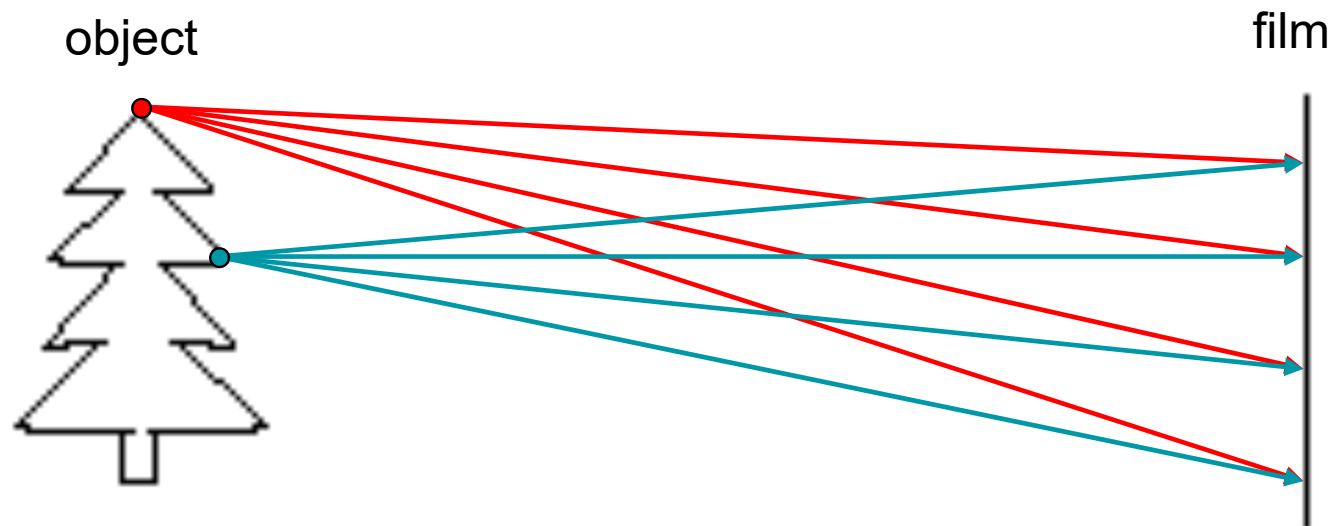
Projection properties

- Parallel lines converge at a vanishing point
 - Each direction in space has its own vanishing point
 - But parallels parallel to the image plane remain parallel



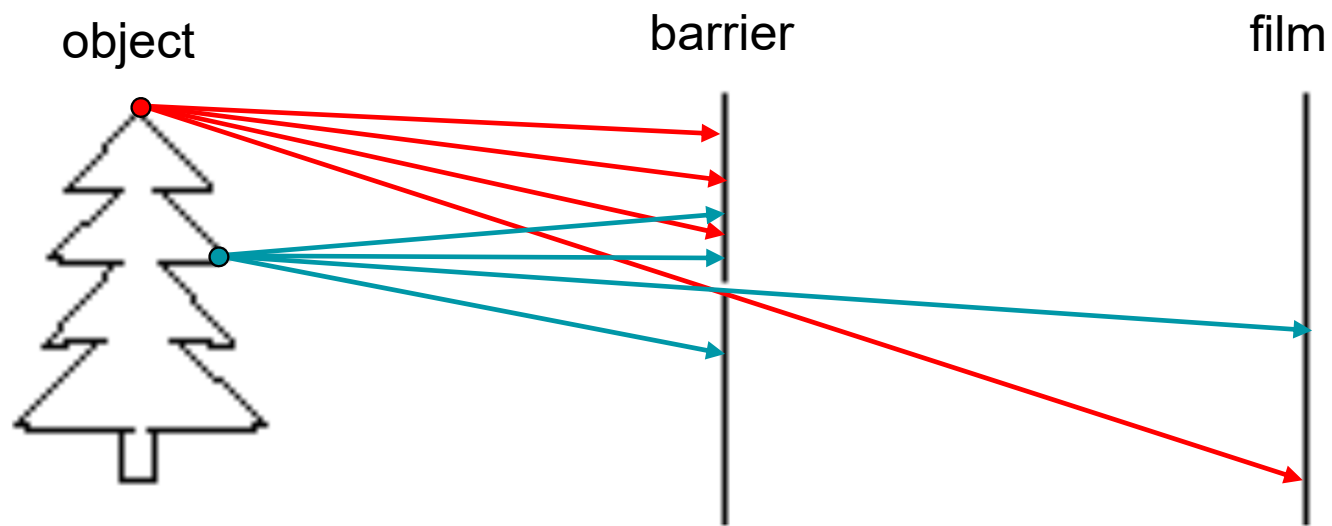
Pinhole camera model

- Design a camera



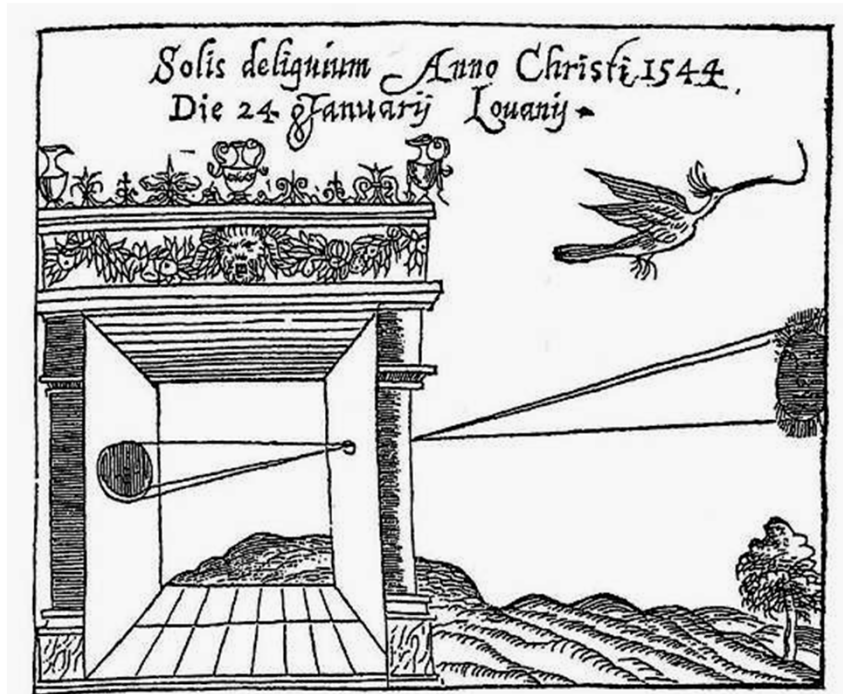
Pinhole camera model

- Design a camera

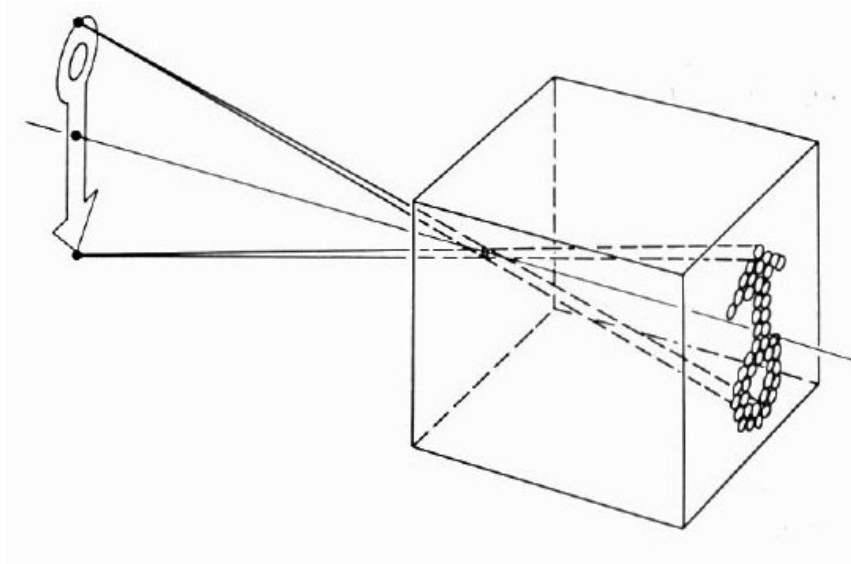


Pinhole camera model

- Design a camera



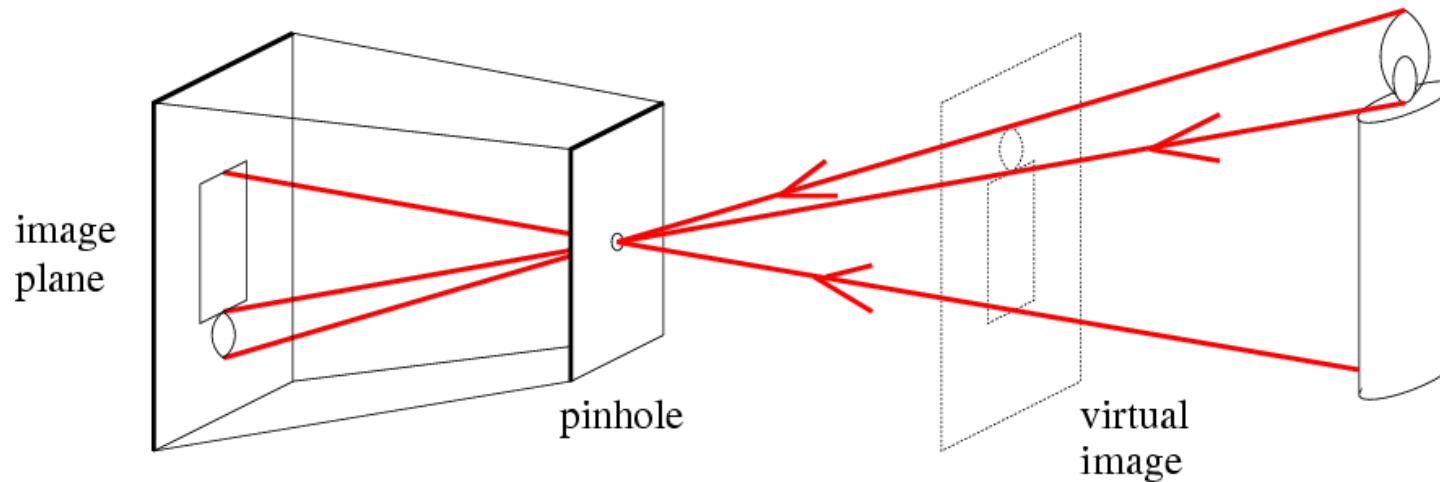
- Basic principle known to Mozi (470-390 BC), Aristotle (384-322 BC)
- Drawing aid for artists: described by Leonardo da Vinci (1452-1519)



Pinhole camera model

■ Pinhole camera

- Box with a small hole in it
- Only one ray from any given point can enter the camera
- When 3D world is projected to 2D, image is inverted



Pinhole camera model

- Design a camera

- Pinhole too big

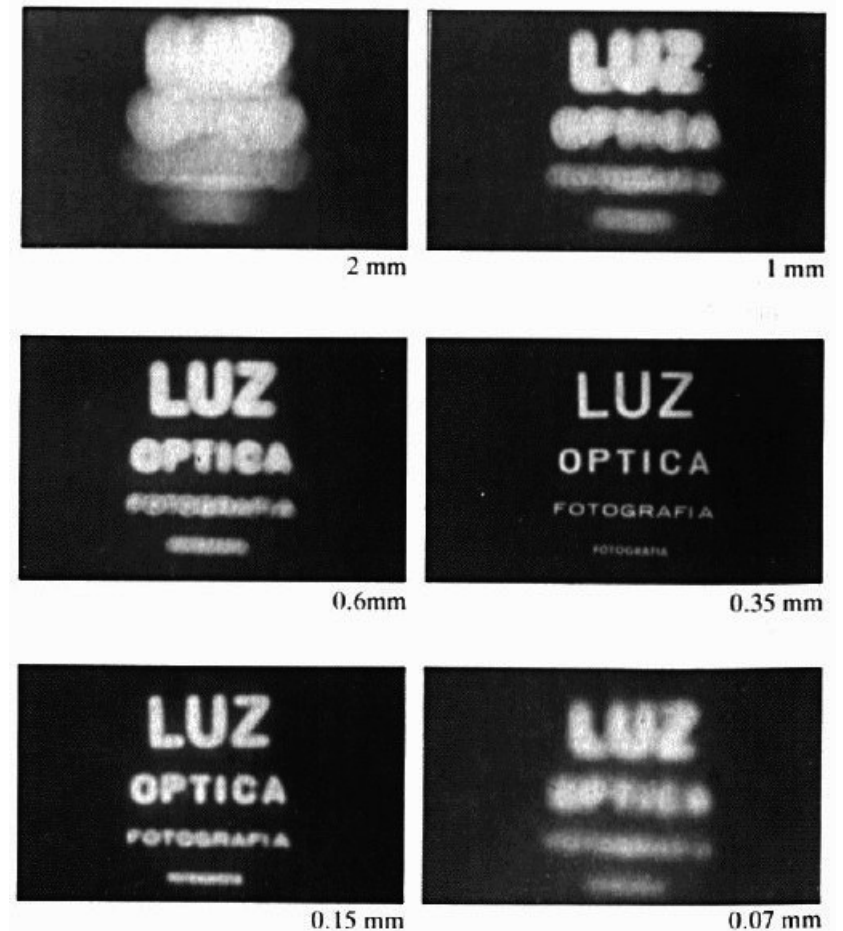
- Many directions are averaged
 - Blurring the image

- Pinhole too small

- Diffraction effects blur the image

- Generally, pinhole cameras are *dark*

- Because very small set of rays from a particular point hits the screen

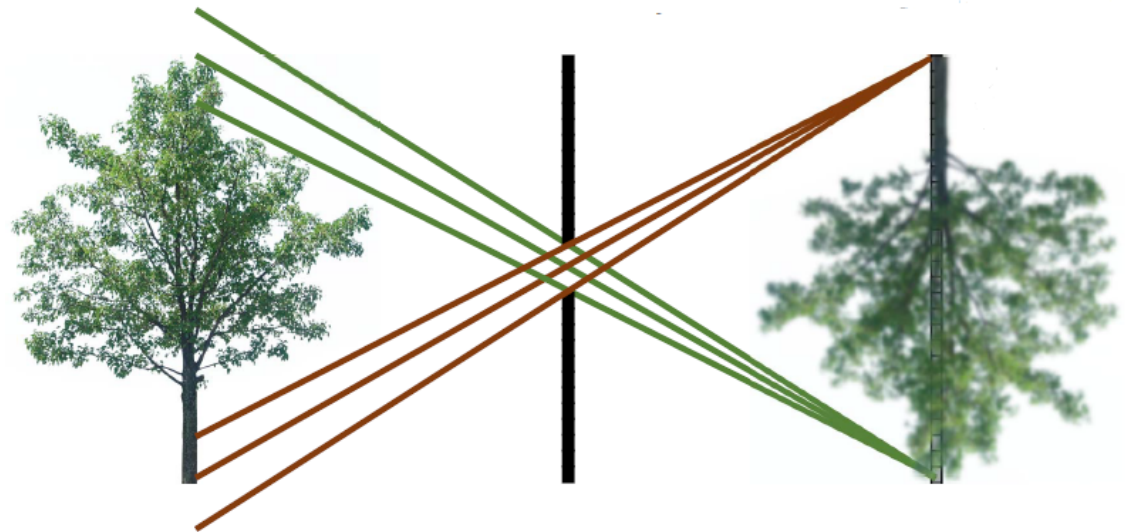


Pinhole camera model

- Design a camera

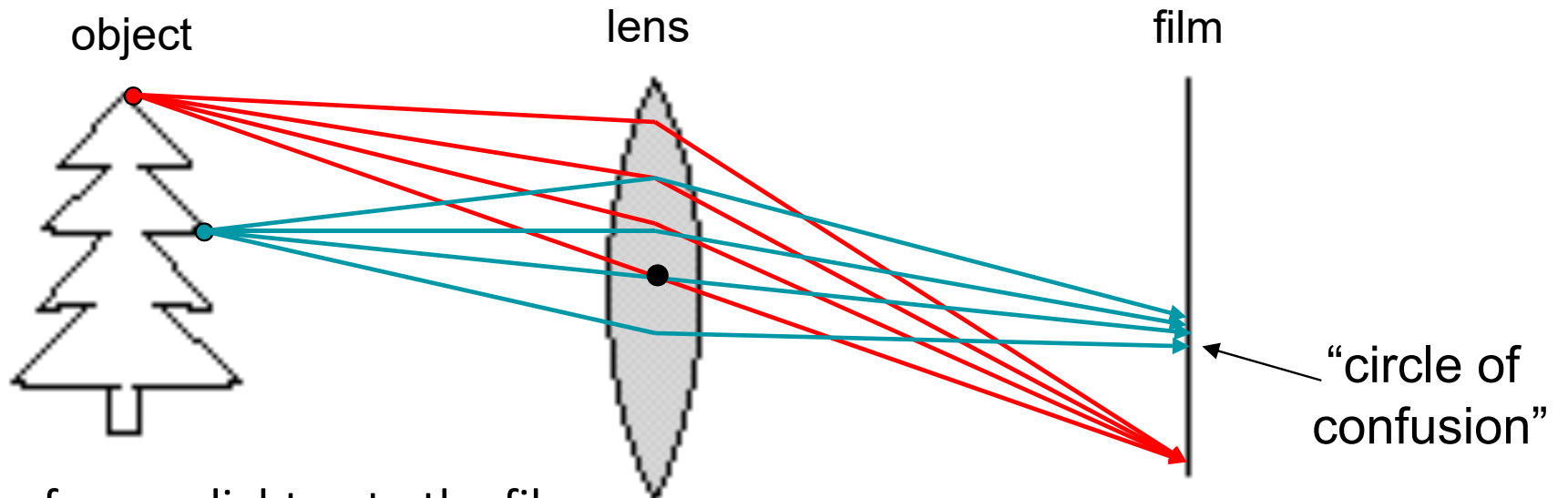
Why not make the aperture as small as possible?

- Less light gets through
- *Diffraction* effects...



Cameras with lenses

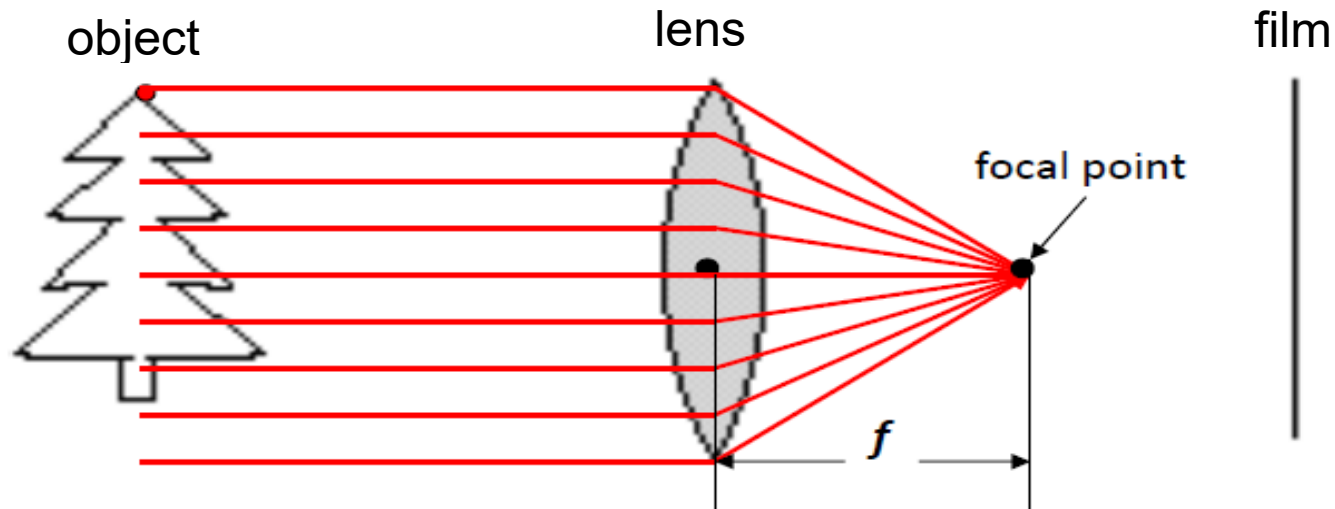
- Design a camera - Adding a lens



- A lens focuses light onto the film
 - There is a specific distance at which objects are “in focus”
 - other points project to a “circle of confusion” in the image
 - Changing the shape of the lens changes this distance

Cameras with lenses

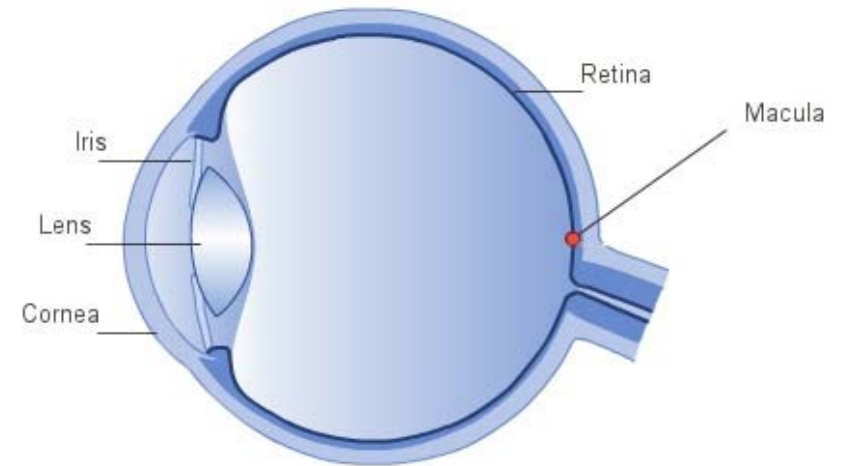
- Design a camera - Adding a lens



- Rays passing through the center are not deviated (pinhole projection model still holds)
- All rays parallel to the optical axis pass through the focal point
- All parallel rays converge to points on the focal plane

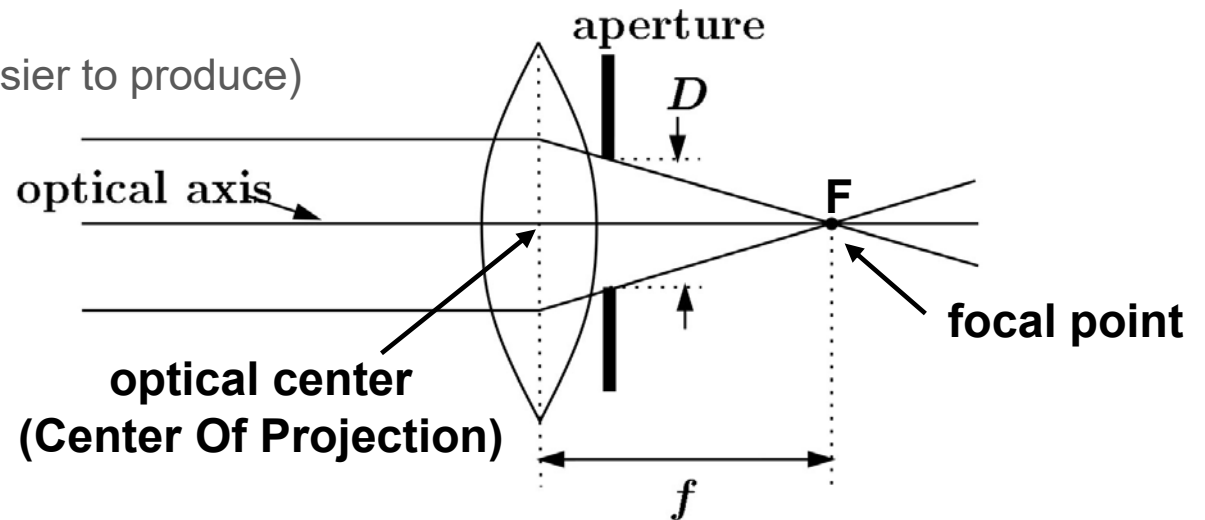
Cameras with lenses

- Human eye is a lens
- Iris
 - Control amount of light passing through lens
- Retina
 - Contains sensor cells, where image is formed
- Fovea
 - Highest concentration of cones



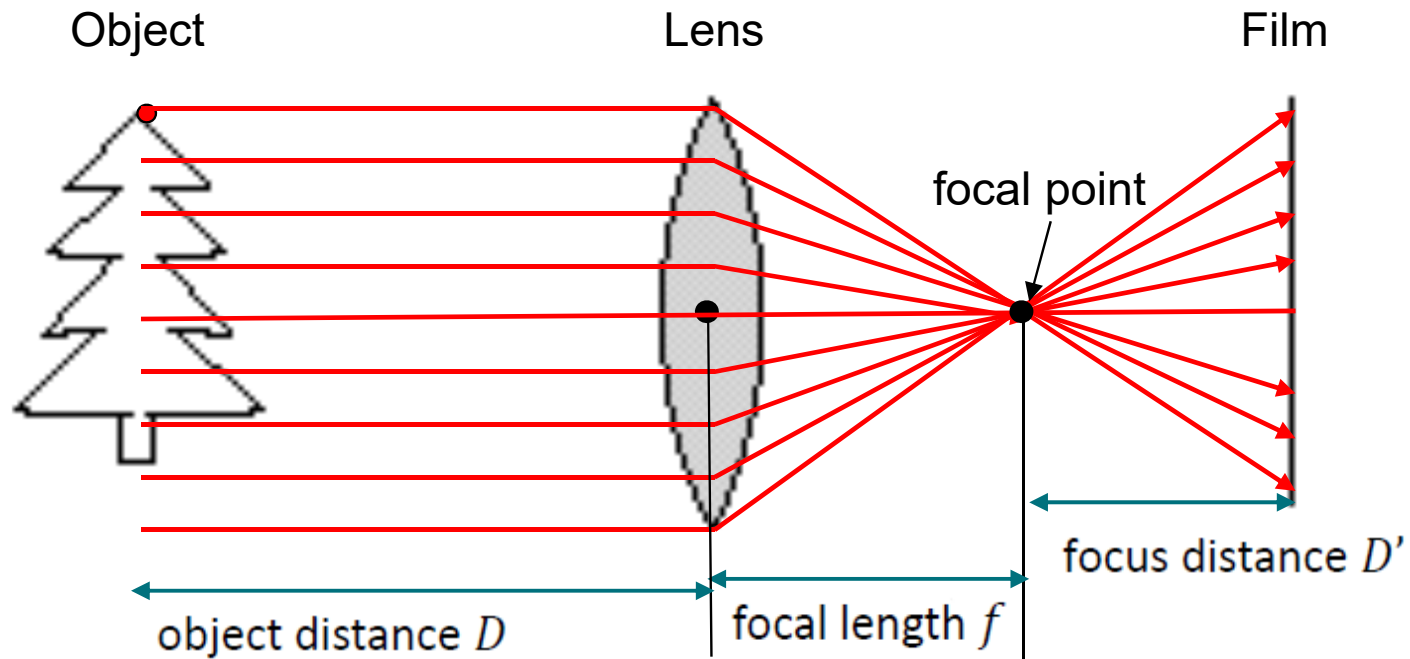
Cameras with lenses

- A lens focuses parallel rays onto a single focal point
 - focal point at a distance f beyond the plane of the lens (the *focal length*)
 - f is a function of the shape and index of refraction of the lens
 - Aperture restricts the range of rays
 - aperture may be on either side of the lens
 - Lenses are typically spherical (easier to produce)



Cameras with lenses

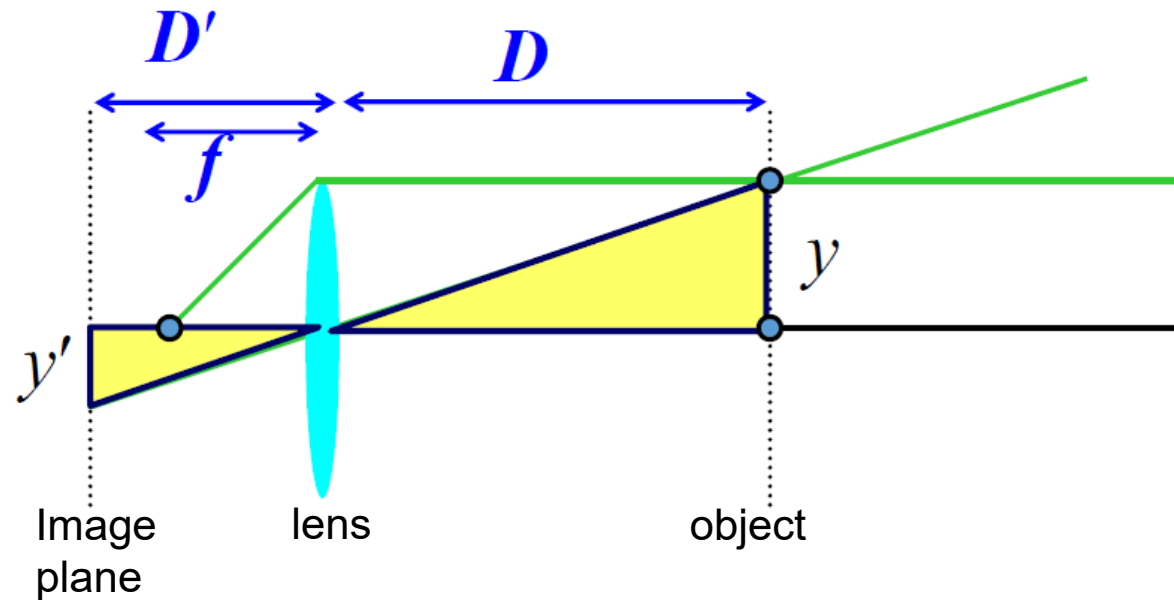
- Focal Length



Cameras with lenses

- Thin Lens Formula

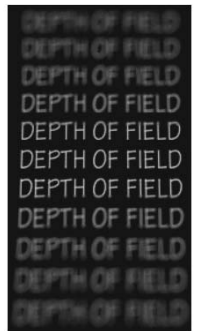
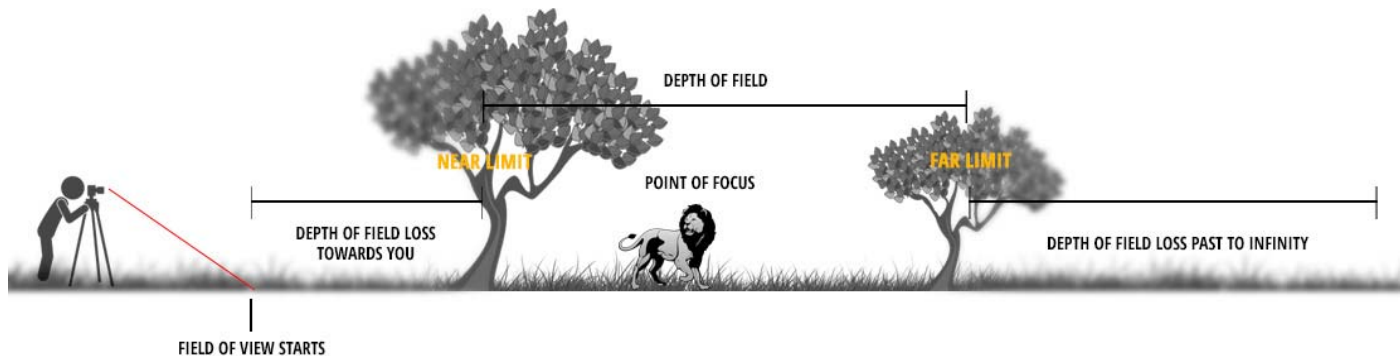
- Any object point satisfying the thin lens equation is in focus
- If the focal plane D' is moved closer, circle of confusion



$$\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}$$

Depth of Field

- In real camera lenses, there is a range of points which are brought into focus at the same distance
- **Depth of field** of the lens , as D gets large(∞) D' approaches f
- Human eye – power of accommodation – changing f



Depth of Field

- How can we control the depth of field?
 - *Changing the aperture size affects depth of field*
 - *A smaller aperture increases the range in which the object is approximately in focus*
 - *But small aperture reduces amount of light need to increase exposure*



Large aperture = small DOF

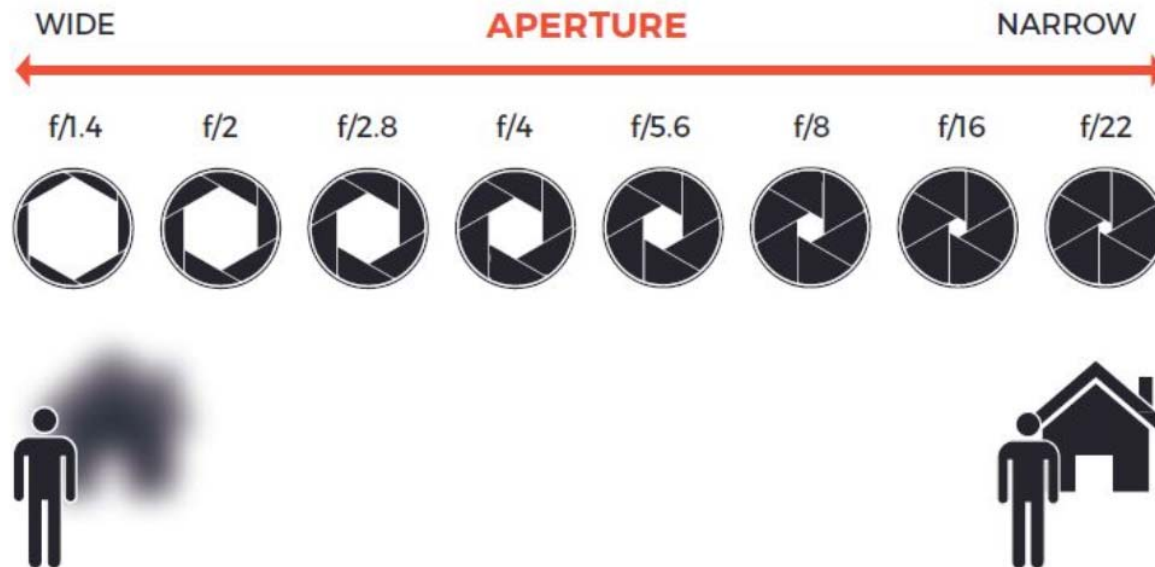


Small aperture = large DOF

Depth of Field

- Aperture

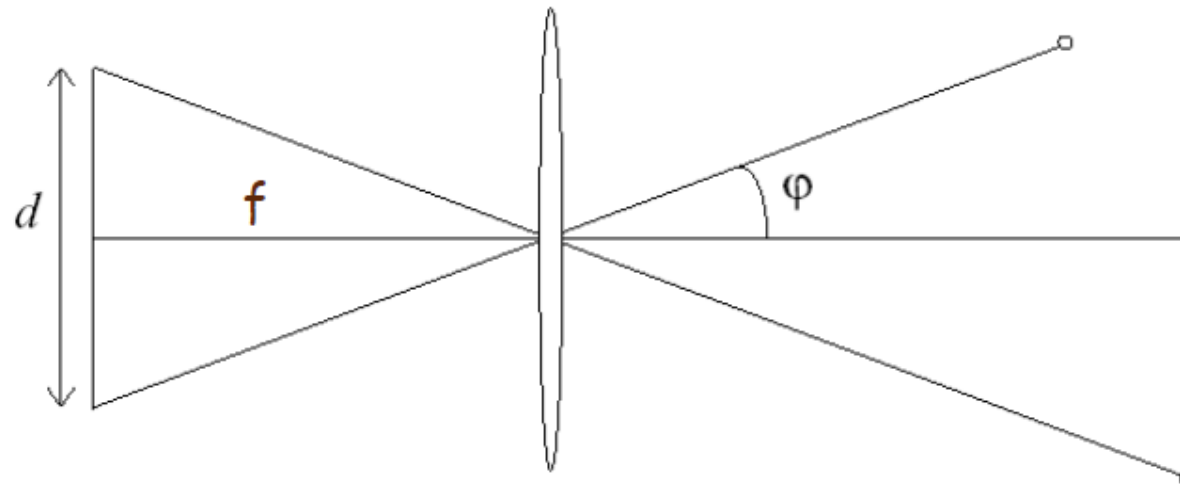
- Smaller aperture increases the range that the object is approximately in focus



Field of View

- FOV depends on focal length and size of the camera retina

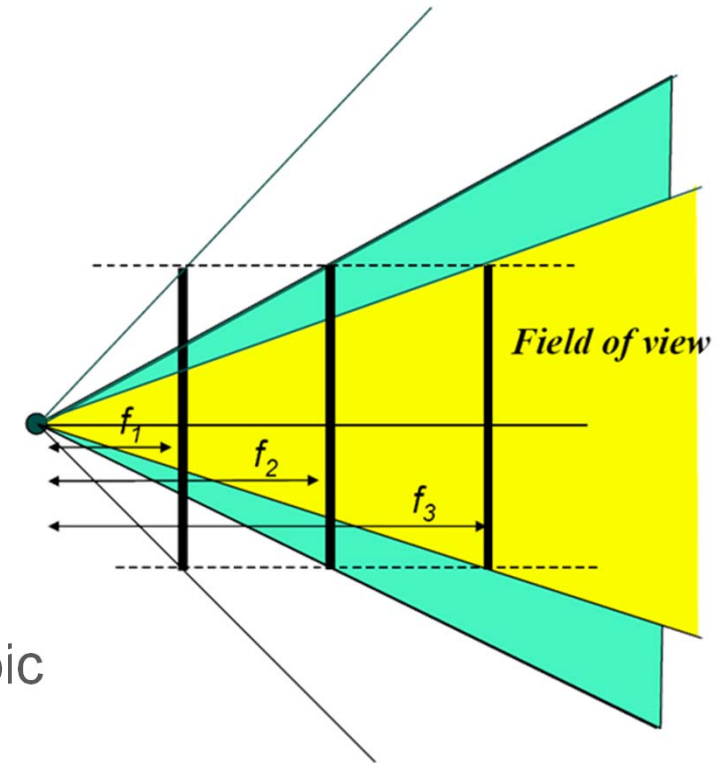
$$\varphi = \tan^{-1}\left(\frac{d}{2f}\right)$$



larger Focal Length $\Rightarrow$ Smaller FOV

Field of View

- Field of view depends on focal length (f)
- As f gets smaller, image becomes wider angle
 - More world points project onto finite image plane
- As f gets larger, image becomes more telescopic
 - Small part of the world projects onto the finite image plane



Field of View

- Angular measure of portion of 3D space seen by the camera



28 mm lens, $65.5^\circ \times 46.4^\circ$



50 mm lens, $39.6^\circ \times 27.0^\circ$



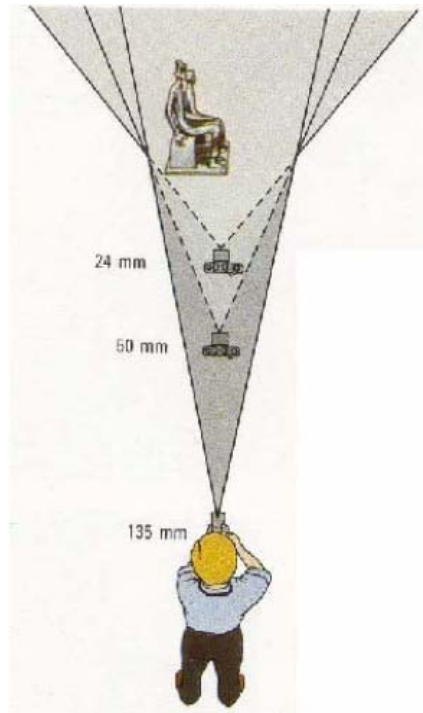
70 mm lens, $28.9^\circ \times 19.5^\circ$



210 mm lens, $9.8^\circ \times 6.5^\circ$

Field of View

- Focal length vs. Viewpoint
 - Telephoto makes it easier to select background
(a small change in viewpoint is a big change in background)



Grand-angle 24 mm



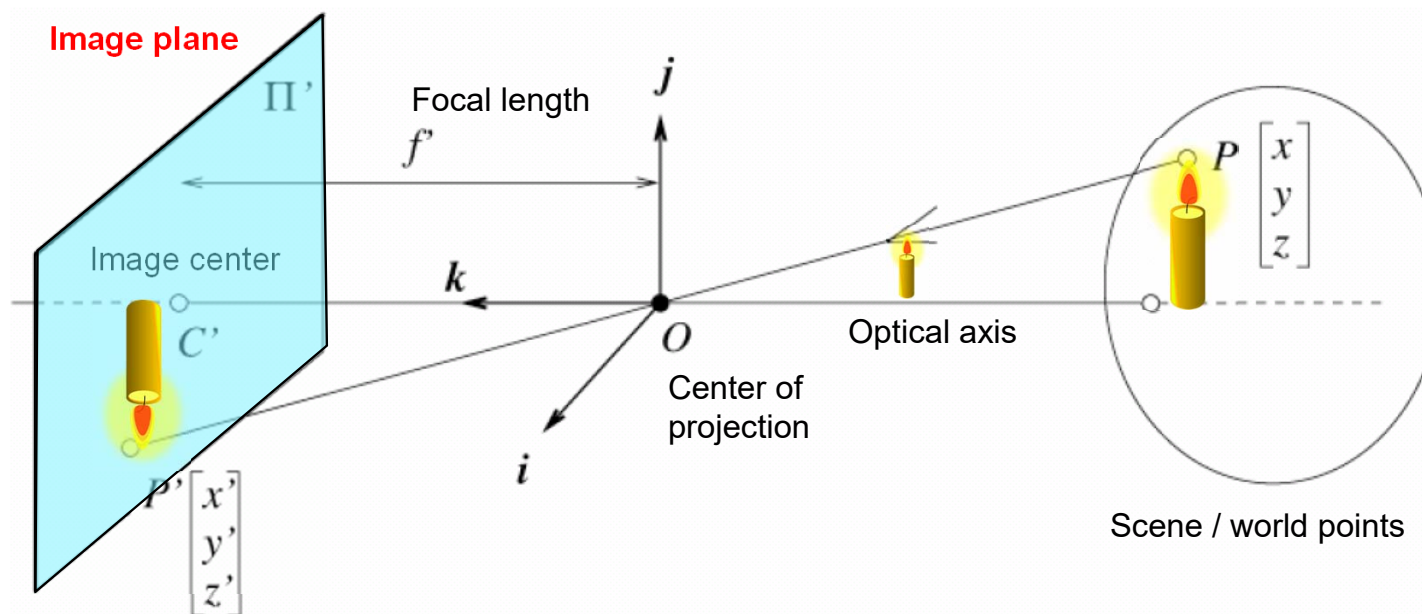
Normal 50 mm



Longue focale 135 mm

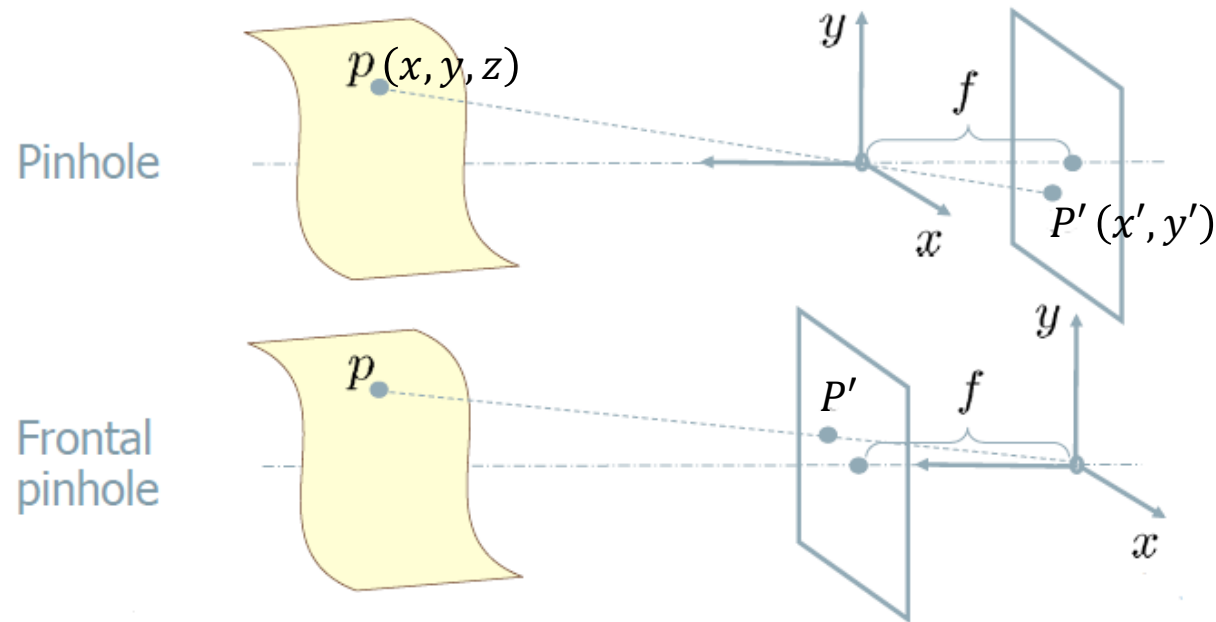
Pinhole Camera Model

- 3D world is mapped to 2D image



Pinhole Camera Model

- Rearranged Pinhole Camera Model



$$(x, y, z) \rightarrow \left(f \frac{x}{z}, f \frac{y}{z}\right)$$

Scene point $\rightarrow$ Image coordinates

Pinhole Camera Model

- What is the camera matrix P for a pinhole camera?

$$x = PX$$

$$P = \begin{bmatrix} ? & ? & ? & ? \\ ? & ? & ? & ? \\ ? & ? & ? & ? \end{bmatrix}$$

Pinhole Camera Model

- Relationship between coordinates in the camera frame and sensor plane

$$P = \begin{bmatrix} ? & ? & ? & ? \\ ? & ? & ? & ? \\ ? & ? & ? & ? \end{bmatrix}$$



$$P = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$(x, y, z) \rightarrow \left(f \frac{x}{z}, f \frac{y}{z}\right)$$



$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \frac{1}{z} \begin{bmatrix} fx \\ fy \\ Z \end{bmatrix} \rightarrow Z \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} fx \\ fy \\ Z \end{bmatrix}$$

- Rewrite by separating the camera parameters and the projection

$$P = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$



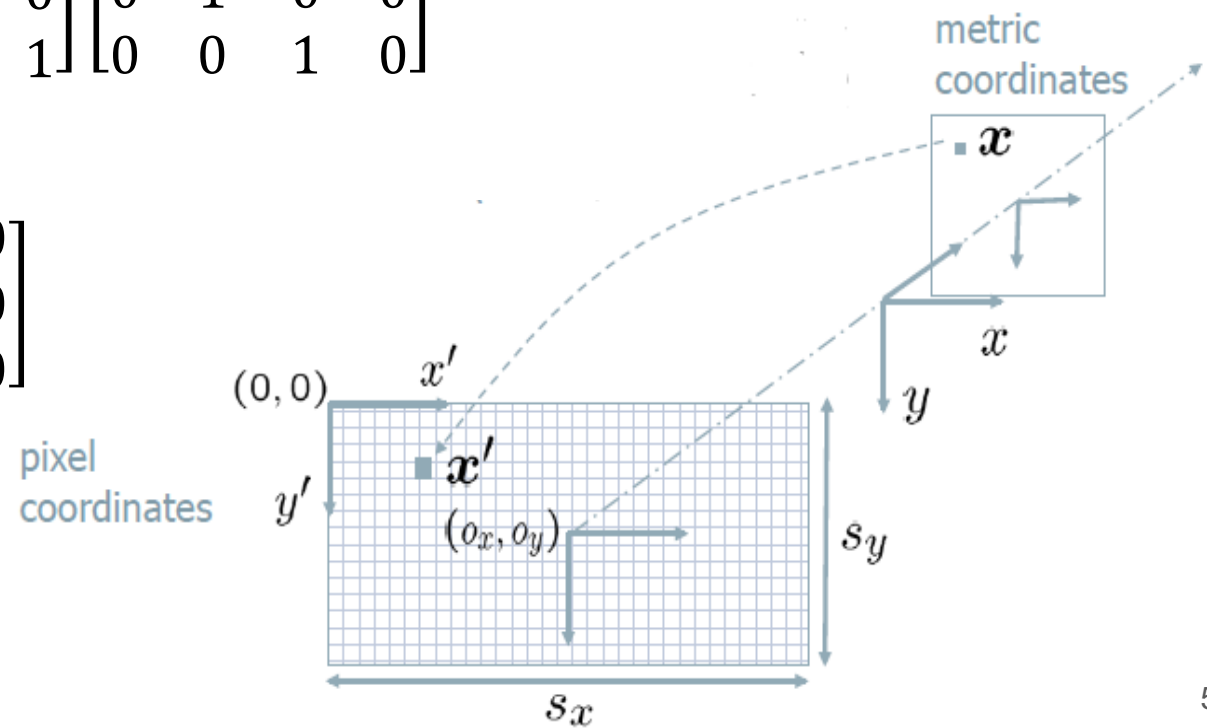
$$P = \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Pinhole Camera Model

- Relationship between coordinates in the sensor plane and image

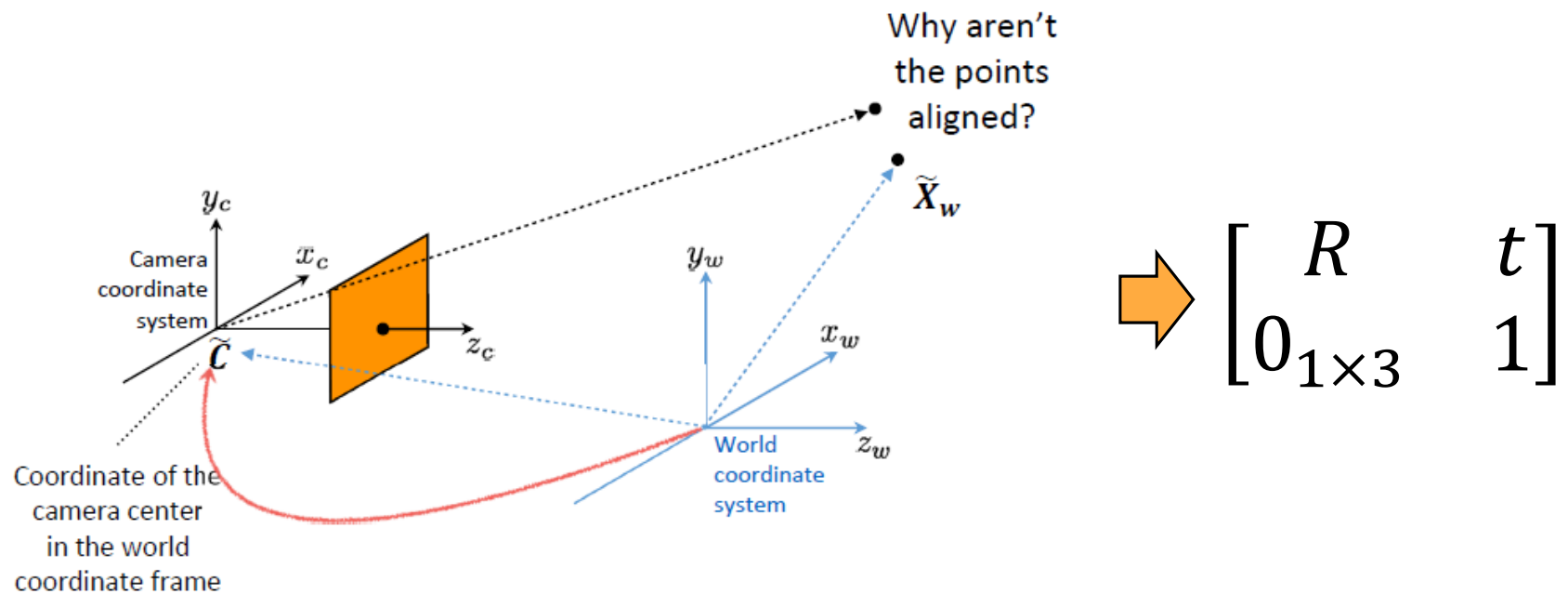
$$P = \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$


$$P = \begin{bmatrix} f & 0 & o_x \\ 0 & f & o_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$



Pinhole Camera Model

- Relationship between coordinates in the world frame and image



Pinhole Camera Model

- Complete Camera Model

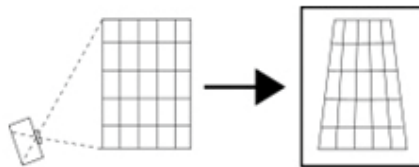
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} f & 0 & o_x \\ 0 & f & o_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{\text{Intrinsic parameters}} \underbrace{\begin{bmatrix} R & t \\ 0_{1 \times 3} & 1 \end{bmatrix}}_{\text{Extrinsic parameters}} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

- More General Camera Model

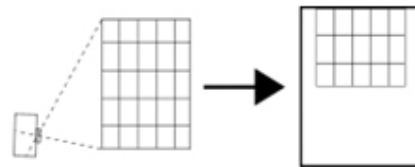
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} f s_x & f s_\theta & o_x \\ 0 & f s_y & o_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} R & t \\ 0_{1 \times 3} & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Applications

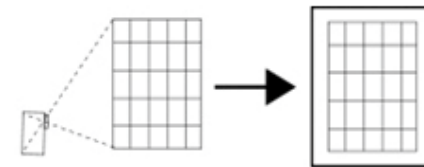
- Perspective distortion
 - Problem for architectural photography
: converging verticals



Tilting the camera upwards results in converging verticals



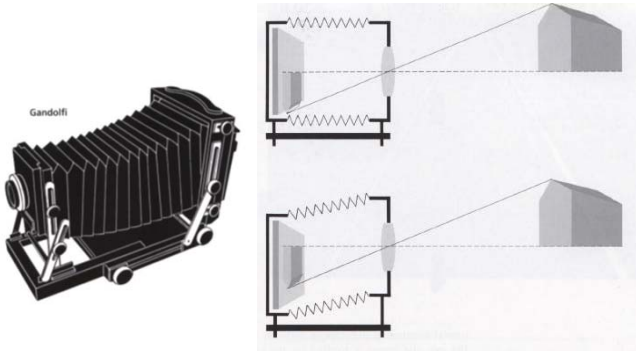
Keeping the camera level, with an ordinary lens, captures only the bottom portion of the building



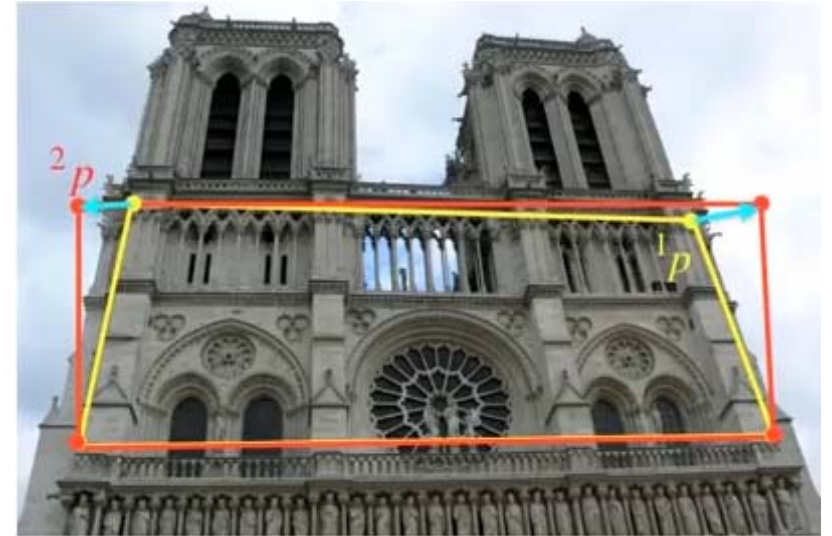
Shifting the lens upwards results in a picture of the entire subject

Applications

- Perspective distortion
 - Problem for architectural photography
: converging verticals
 - Solution : Lens shifted
or Perspective rectification(modeling distortion)

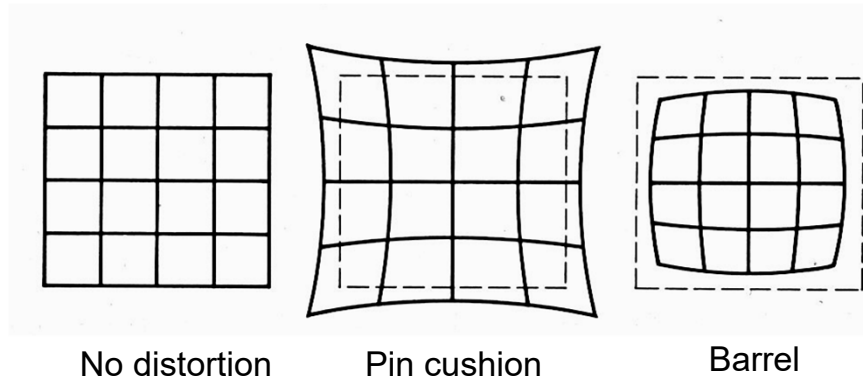


$${}^2p = \mathbf{H} {}^1p$$



Applications

- Radial distortion
 - Caused by imperfect lenses
 - Deviations are most noticeable for rays that pass through the edge of the lens



Thanks!